

Database Schema

Behavior-Aware Portfolio Advisor

Design Principles

- **History over overwrite:** new questionnaire submissions or retakes create new rows rather than updating existing ones, so past results aren't lost.
- **JSONB for flexible data:** raw answers, portfolio weights, and simulation percentiles are stored as JSONB so the questionnaire or asset list can change later without a schema migration.
- **Chained references:** each table points back to the one that produced it (financial/behavioral data → risk scores → portfolio → simulation), so any result can be traced back through the pipeline.
- **Range buckets + derived numbers:** income, goal amount, and goal timeframe are collected as friendly range buckets in the UI, but each stores a derived numeric estimate alongside it for the model to actually use.

Entity Relationship Diagram

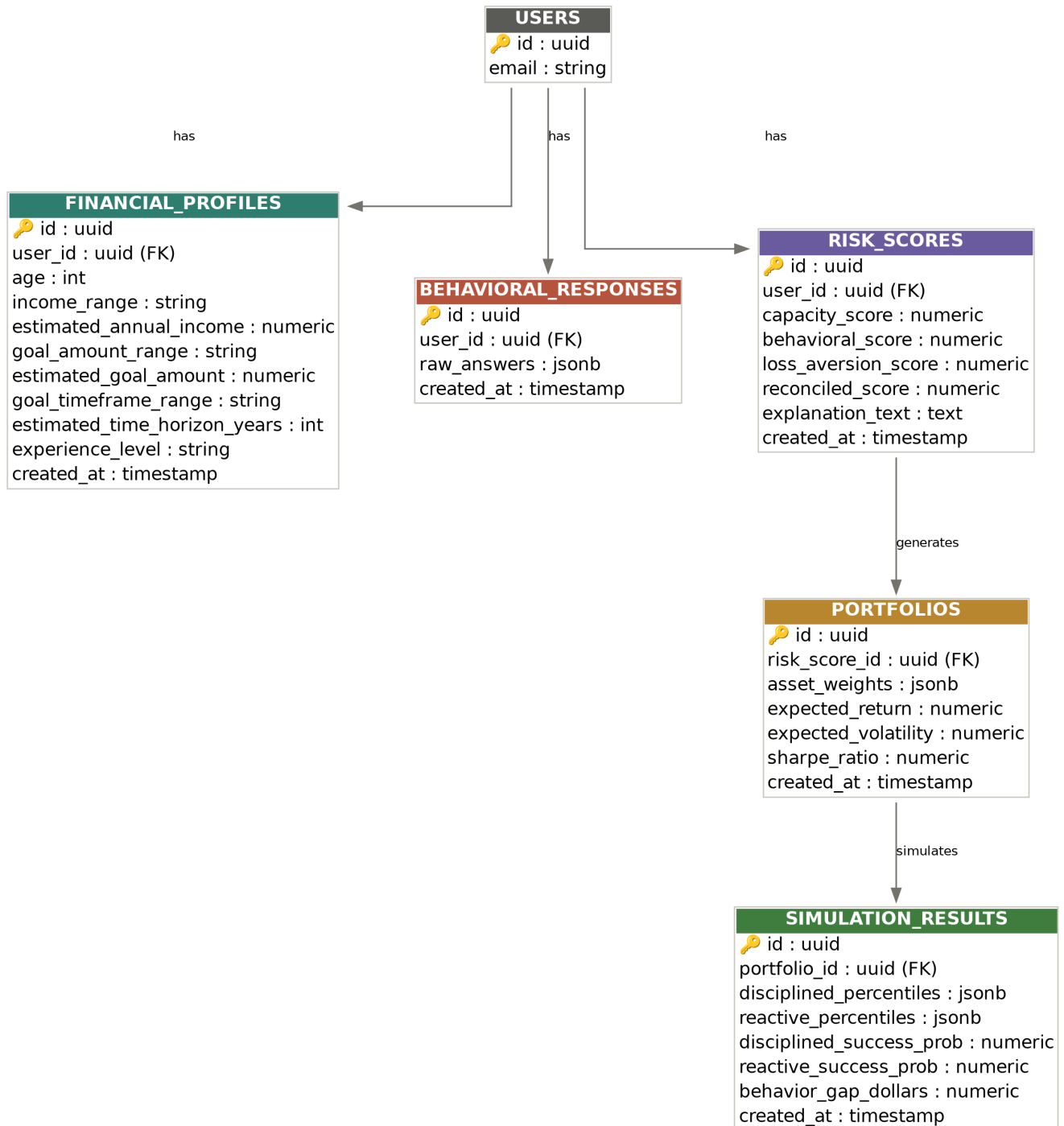


Table Descriptions

users

Managed mostly by Supabase Auth.

Column	Type	Notes
id	uuid (PK)	
email	string	

financial_profiles

Column	Type	Notes
id	uuid (PK)	
user_id	uuid (FK)	references users
age	int	
income_range	string	e.g. "\$50,000–\$75,000"
estimated_annual_income	numeric	derived midpoint of selected range
goal_amount_range	string	e.g. "\$100,000–\$500,000"
estimated_goal_amount	numeric	derived midpoint of selected range
goal_timeframe_range	string	e.g. "10–20 years"
estimated_time_horizon_years	int	derived midpoint of selected range
experience_level	string	none / some / experienced
created_at	timestamp	

behavioral_responses

Column	Type	Notes
id	uuid (PK)	
user_id	uuid (FK)	references users
raw_answers	jsonb	all questionnaire answers
created_at	timestamp	

risk_scores

Column	Type	Notes
id	uuid (PK)	
user_id	uuid (FK)	references users
capacity_score	numeric	from the ML risk capacity model
behavioral_score	numeric	from the behavioral questionnaire
loss_aversion_score	numeric	

reconciled_score	numeric	blended final score used for portfolio construction
explanation_text	text	the “tension” explanation when scores diverge
created_at	timestamp	

portfolios

Column	Type	Notes
id	uuid (PK)	
risk_score_id	uuid (FK)	references risk_scores
asset_weights	jsonb	e.g. {"VTI": 0.4, "VXUS": 0.2, "BND": 0.3, "VNQ": 0.1}
expected_return	numeric	
expected_volatility	numeric	
sharpe_ratio	numeric	
created_at	timestamp	

simulation_results

Column	Type	Notes
id	uuid (PK)	
portfolio_id	uuid (FK)	references portfolios
disciplined_percentiles	jsonb	10th/50th/90th percentile outcomes
reactive_percentiles	jsonb	same, for the behavior-adjusted path
disciplined_success_prob	numeric	probability of reaching the goal
reactive_success_prob	numeric	same, for the behavior-adjusted path
behavior_gap_dollars	numeric	dollar difference between the two
created_at	timestamp	

Bucketed Field Options

Annual income

- Less than \$30,000
- \$30,000 – \$50,000
- \$50,000 – \$75,000
- \$75,000 – \$100,000
- \$100,000 – \$150,000
- \$150,000+

Goal amount

- Less than \$10,000
- \$10,000 – \$50,000
- \$50,000 – \$100,000
- \$100,000 – \$500,000
- \$500,000+

Goal timeframe

- Less than 5 years
- 5–10 years
- 10–20 years
- 20+ years

Each bucket should map to a single representative number (e.g. midpoint) in code — this is what gets stored in the *estimated_** columns and passed into the models.